

Saroj Khadka

Graduate Researcher

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Summary

Biomedical Science PhD candidate with strong research experience in bacterial virulence mechanisms, host-pathogen interactions, and microbial gene regulation. Skilled in molecular biology and microbiological techniques, mentoring and science communication. Motivated to build an independent career in biomedical research.

Education

University of Pittsburgh, Microbiology and Immunology – Pittsburgh, PA, USA 2024 – 2026

- {‘Thesis’: ‘Mechanism and Regulation of *Klebsiella pneumoniae* Capsular Polysaccharide Chain Length and Attachment.’}
- {‘Advisor’: ‘Dr. Laura A. Mike’}

Tribhuvan University, Medical Microbiology – Kathmandu, Nepal 2017 – 2019

- {‘Thesis’: ‘Susceptibility to Fluoroquinolones and *gyrA* ser83 Mutation in *Salmonella enterica* serovar Typhi at a Referral Hospital of Kathmandu Valley.’}
- {‘Advisor’: ‘Dr. Megharaj Banjara.’}

Experience

Division of Infectious Diseases, Department of Medicine, Graduate Student Researcher – 2024 – 2026
University of Pittsburgh, PA

Teaching at Palmer Physical Laboratory (now 302 Frist Campus Center). While not a professor at Princeton, I associated with the physics professors and continued to give lectures on campus.

- Relativity

B-Sure Path Lab & Diagnostic Centre, Molecular Biologist – Biratnagar, Nepal 2020 – 2021

- Description 1.

Volunteer

People’s Climate March, Lead Organizer – Zurich, Switzerland Apr 2014 – July 2015

Lead organizer for the New York City branch of the People’s Climate March, the largest climate march in history.

- Awarded ‘Climate Hero’ award by Greenpeace for my efforts organizing the march.
- Men of the year 2014 by Time magazine

Awards

AHA Predoctoral Fellow Jan 2022

The Nobel Prizes are five separate prizes that, according to Alfred Nobel’s will of 1895, are awarded to ‘those who, during the preceding year, have conferred the greatest benefit to humankind.’

American Heart Association

Graduate Student Researcher 2029

Awarded for outstanding scientific achievement
German Physical Society

Publications

Zur Elektrodynamik bewegter Körper

It concerned an interpretation of the Michelson–Morley experiment and the properties of light and time. Special relativity incorporates the principle that the speed of light is the same for all inertial observers regardless of the state of motion of the source.

Albert Einstein

en.wikisource.org/wiki/Translation:On_the_Electrodynamics_of_Moving_Bodies

Über einen die Erzeugung und Verwandlung des Lichtes betreffenden heuristischen Gesichtspunkt

In the second paper, he applied the quantum theory to light to explain the photoelectric effect. In particular, he used the idea of light quanta (photons) to explain experimental results, but stressed the importance of the experimental results. The importance of his work on the photoelectric effect earned him the Nobel Prize in Physics in 1921.

Albert Einstein

de.wikisource.org/

wiki/%C3%9Cber_einen_die_Erzeugung_und_Verwandlung_des_Lichtes_betreffenden_heuristischen_Gesichtspunkt

Die Grundlage der allgemeinen Relativitätstheorie

The publication of the theory of general relativity made him internationally famous. He was professor of physics at the universities of Zurich (1909–1911) and Prague (1911–1912), before he returned to ETH Zurich (1912–1914).

Albert Einstein

de.wikisource.org/wiki/Die_Grundlage_der_allgemeinen_Relativit%C3%A4tstheorie

Skills

Physics

Languages

German

Native speaker

English

Fluent

Interests

Physics

Certificates

Machine Learning

Jan 2018

Quantum Computing

Jan 2018

Quantum Information

Jan 2018

Projects

Quantum Computing

Jan 2018 – Jan 2018

Quantum computing is the use of quantum-mechanical phenomena such as superposition and entanglement to perform computation. Computers that perform quantum computations are known as quantum computers.

- Quantum Teleportation
- Quantum Cryptography

References

Professor John Doe

Professor Jane Smith